

AI STUDY BUDDY

DOCUMENTATION

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2. Problem Statement & Proposed Solution

AI StudyBuddy

AI Assistant for Smarter Learning
and Study Support

Problem Statement

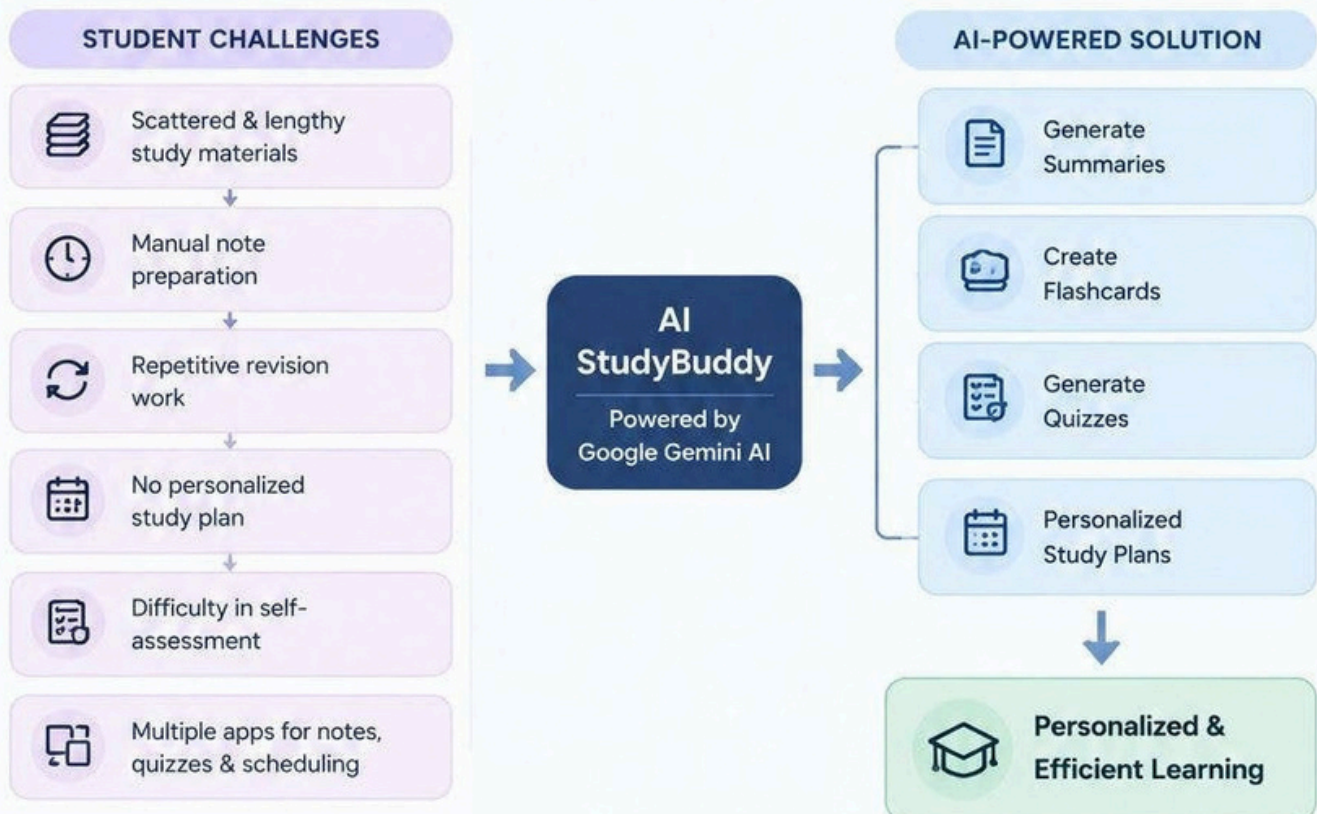
Students often face difficulties while preparing for examinations due to:

- Lengthy and scattered study materials
- Time-consuming manual note preparation
- Repetitive creation of flashcards and revision questions
- Lack of personalized study plans
- Difficulty evaluating knowledge through self-generated quizzes
- Using multiple applications for notes, quizzes and scheduling

Proposed Solution

AI StudyBuddy provides a centralized AI-powered learning platform that automates these study activities. Students can upload their study materials and use Google Gemini AI to generate summaries, flashcards, quizzes and personalized study plans.

Problem → Solution Flow



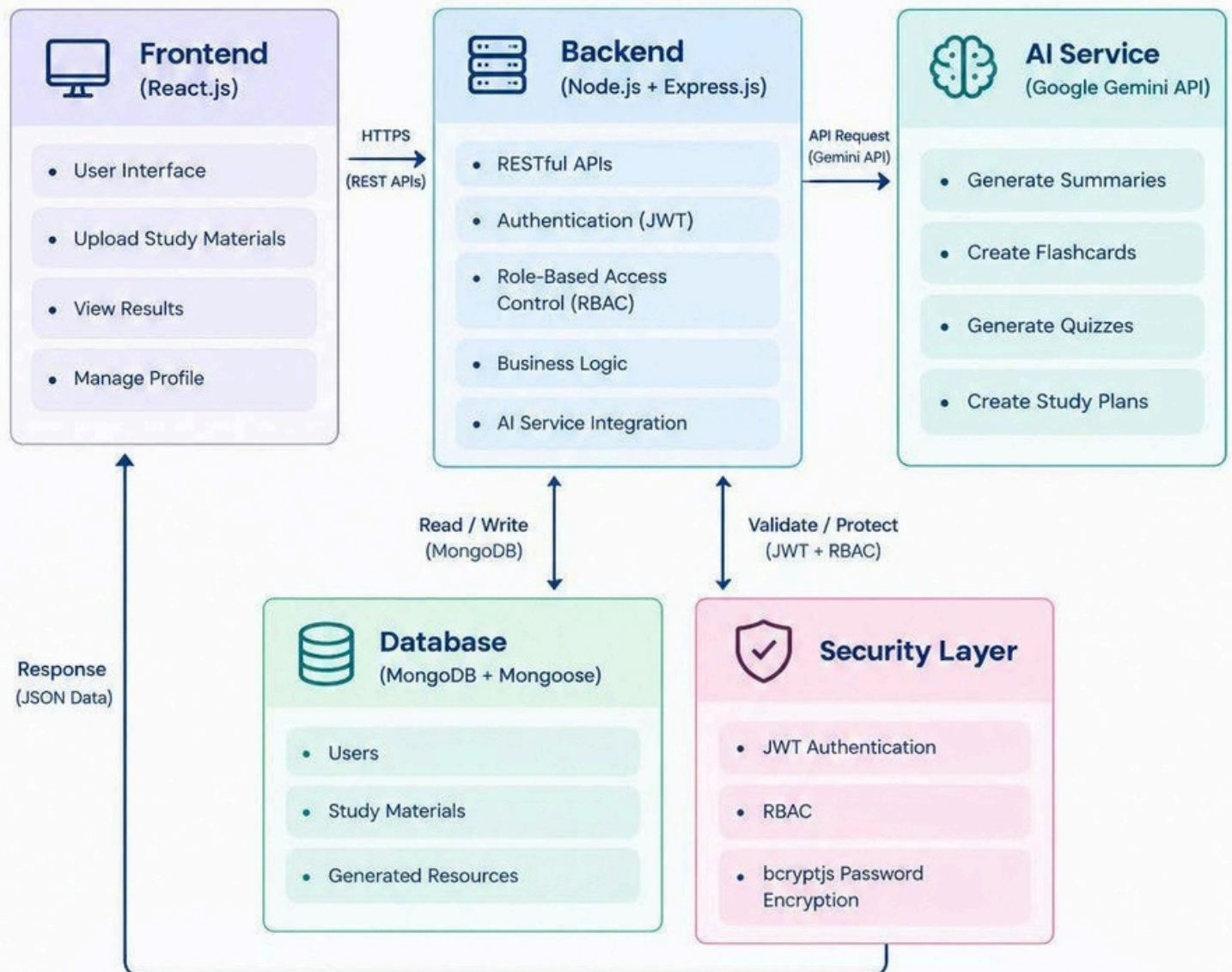
Key Objective

To reduce manual study preparation and provide students with intelligent, personalized, and centralized learning support.

3.

System Architecture

AI StudyBuddy – AI Assistant for Smarter Learning and Study Support



Architecture Flow

- 1 User interacts with the React frontend.
- 2 Frontend sends requests to the Node.js + Express.js backend.
- 3 Backend processes the request and communicates with MongoDB and Google Gemini AI API.
- 4 AI service generates the required response.
- 5 Response is sent back to the frontend and displayed to the user.

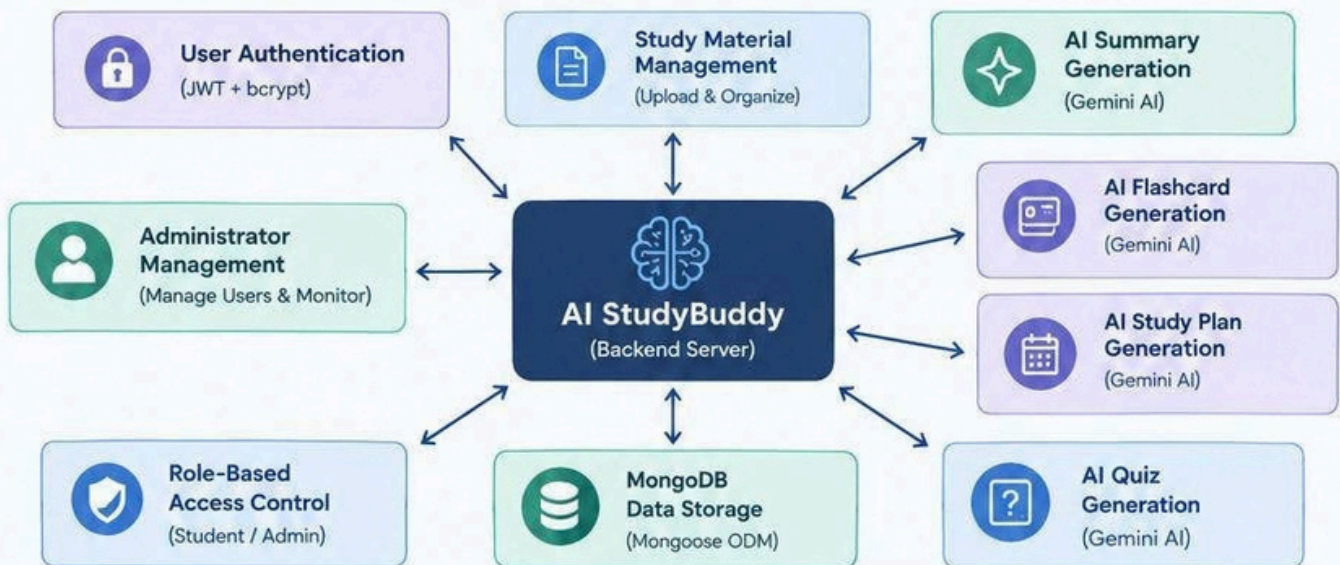
Key Components

- Frontend**
- React.js
- Backend**
- Node.js, Express.js
- Database**
- MongoDB, Mongoose
- AI Service**
- Gemini API
- Security**
- JWT, RBAC, bcryptjs

Benefits of Architecture

- ✓ Scalable and maintainable structure
- ✓ Secure communication between services
- ✓ Modular design
- ✓ Efficient request handling
- ✓ Easy feature integration

Module Relationship Diagram



User Flow – Student



Key Features

- 1 JWT Authentication & bcrypt**
Secure user login with JWT tokens and password encryption using bcrypt.
- 2 Study Material Management**
Upload and manage study materials (PDFs, notes, text, etc.).
- 3 AI Summary Generation**
Generate concise and easy-to-understand summaries using Google Gemini AI.
- 4 AI Flashcard Generation**
Create flashcards for active recall learning using Gemini AI.
- 5 AI Quiz Generation**
Generate multiple-choice quizzes for self-assessment.
- 6 Personalized Study Plans**
Create study plans based on user goals and available study time.
- 7 Role-Based Access Control (RBAC)**
Manage student and admin privileges securely.
- 8 MongoDB Storage**
Store users and generated learning resources efficiently using MongoDB with Mongoose.

Roles

- Student**
- Register and log in to the application
 - Upload study materials
 - Use AI features (summary, flashcards, quizzes, study plans)
 - Manage generated resources (view / save)

- Administrator**
- Manage registered users
 - Monitor application activity and system usage
 - Maintain platform integrity



Database Overview

The database is designed to store and manage all the core data of AI StudyBuddy. It maintains user information and connects related data such as study materials, summaries, flashcards, quizzes and study plans. Proper relationships are set between the tables to ensure data consistency and easy access for personalized learning.



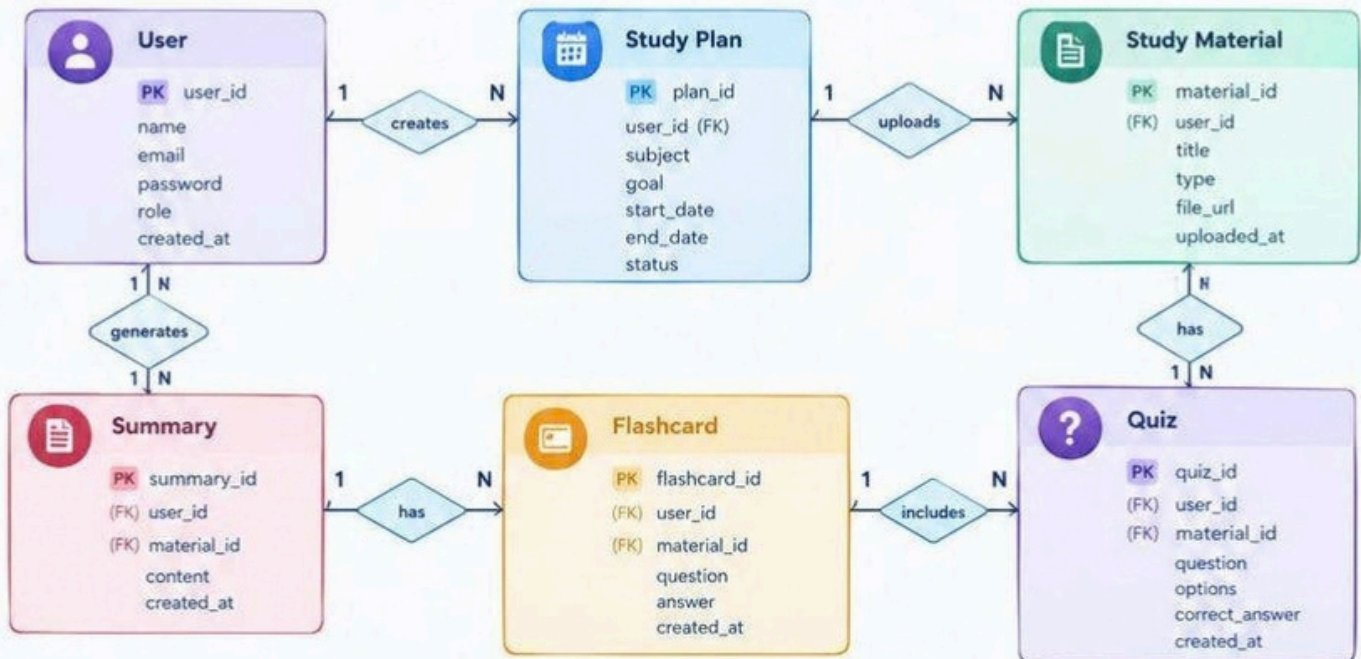
Database Used

MongoDB

- NoSQL database
- Flexible schema
- Stores data in collections
- Scalable and cloud-ready



ER Diagram



Entity Descriptions



User

Stores user account details and authentication information. One user can create multiple study plans, materials, summaries, flashcards and quizzes.



Study Plan

Contains the user's study goals and schedule. Each plan is linked to one user.



Study Material

Stores uploaded PDFs, notes, text or other resources. Each material belongs to one user and can have multiple quizzes, flashcards and summaries.



Summary

Stores AI-generated summaries for each material. Each summary is linked to one user and one material.



Flashcard

Stores flashcards created from a material. Each flashcard is linked to one user and one material.



Quiz

Stores quizzes based on a material. Each quiz is linked to one user and one material.



Relationships Summary

- 1 User 1 —> Study Plan (creates)
- 2 User 1 —> Summary (generates)
- 3 User 1 —> Flashcard (has)
- 4 Study Plan 1 —> Study Material (uploads)
- 5 Study Material 1 —> Quiz (has)
- 6 Flashcard 1 —> Quiz (includes)



Key Benefits

- ✓ Maintains data consistency with clear relationships
- ✓ Supports personalized learning experience
- ✓ Allows easy retrieval and management of study resources
- ✓ Scalable and flexible for future features



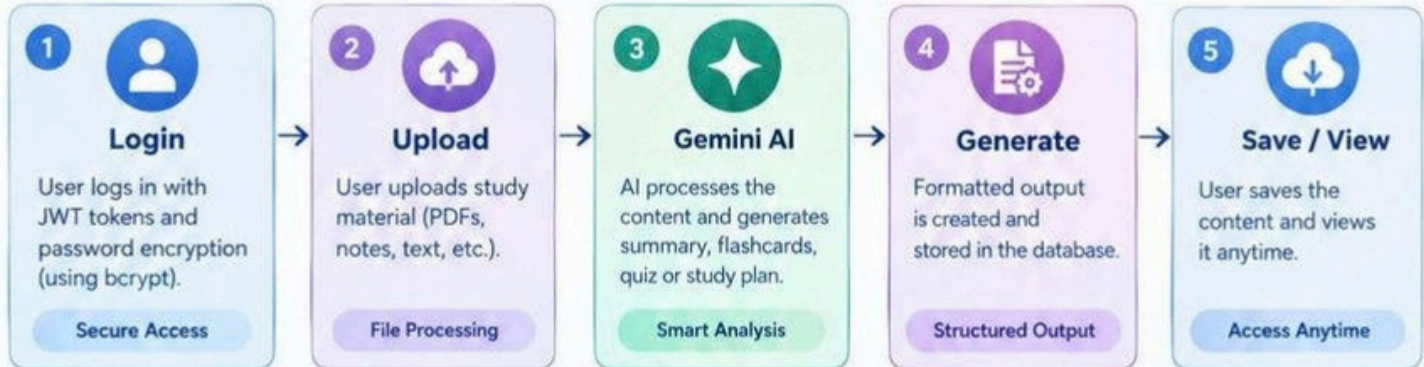
Future Enhancements

- Add more fields for better analytics (e.g., tags, ratings)
- Integrate with real-time collaboration features
- Implement data backup and recovery
- Support multi-language content storage



AI Processing Flow

The user provides input (file or notes), which is processed using Gemini AI to generate structured learning content. The output is then saved and can be viewed anytime in the app.



AI Powers Personalized Learning



AI Model & Tools



Gemini AI (Google Gemini)

- Text generation
- Summarization
- Flashcards & quizzes
- Study plan creation
- Context-aware responses



Tools & Technologies



Google Gemini AI
(AI model)



FastAPI
(Backend API)



MongoDB
(Database)



JWT + bcrypt
(Authentication)



Prompt Example



User Input

"Generate a short summary, 5 flashcards and 2 MCQ questions from this PDF."



Gemini AI Response

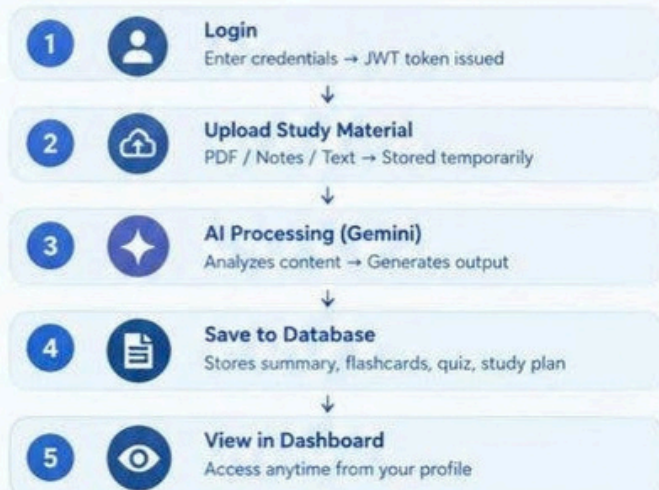
- Concise summary
- 5 flashcards (Q&A format)
- 2 multiple-choice questions
- Structured JSON output



The generated content is formatted and stored in the database for future access.



User Flow – AI Processing



Key Points



Secure

JWT + bcrypt for authentication and data protection.



Smart

Uses Gemini AI for accurate and context-based generation.



Structured

Output is organized and stored in MongoDB.



Fast

Built with FastAPI for quick and reliable responses.



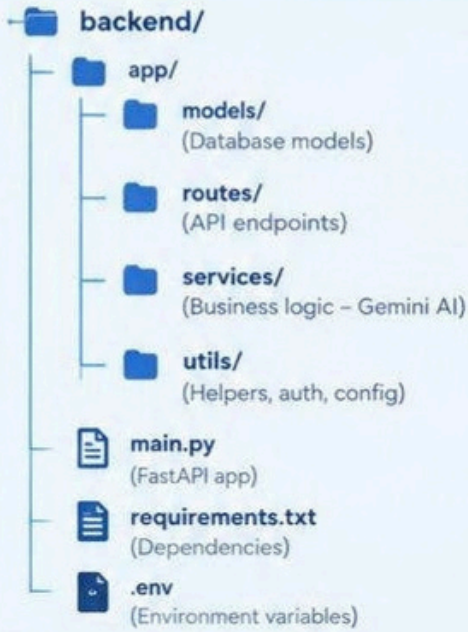
User-Centric

Designed for easy access, learning and progress tracking.

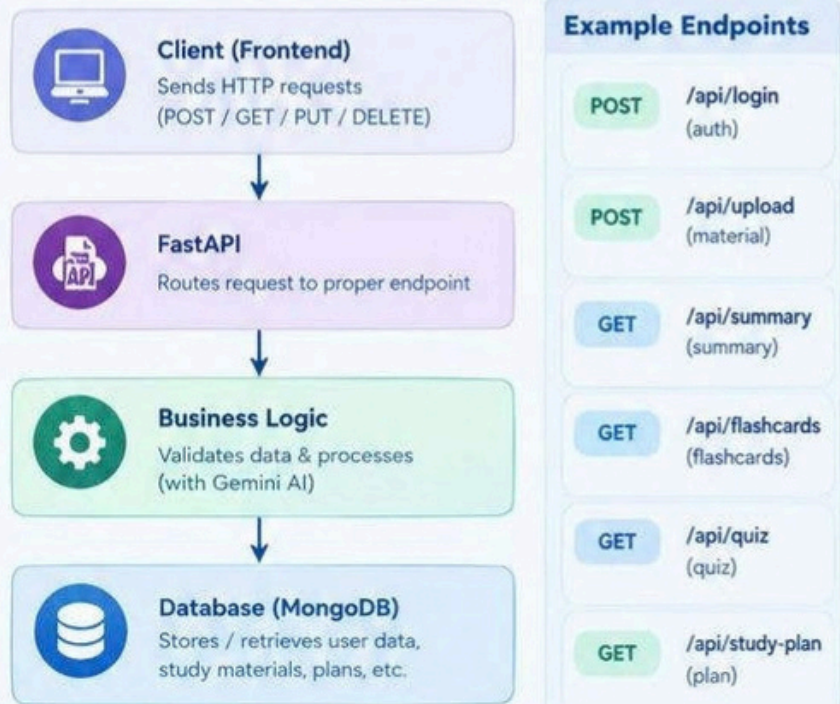


Backend Structure

The backend is built using FastAPI (Python) and connects with MongoDB. It handles authentication, file processing, AI integration and data management.



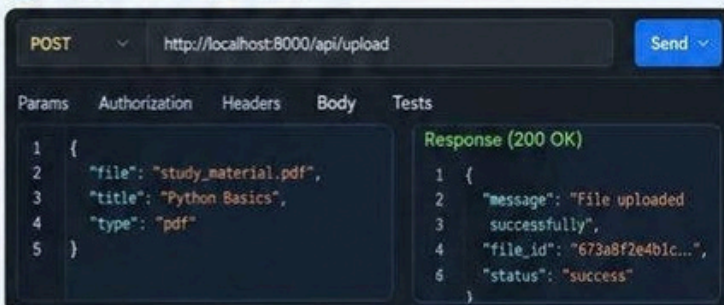
REST API Flow



Code & API Testing

After building the backend, APIs are tested using tools like Postman to ensure all endpoints work correctly.

Sample API Request (Postman)



Tested Endpoints

POST	/api/login	User authentication
POST	/api/upload	Upload study material (PDF/Notes/Text)
GET	/api/summary	Get AI generated summary
GET	/api/flashcards	Get flashcards
GET	/api/quiz	Get quiz questions
GET	/api/study-plan	Get structured study plan



Integration & Configuration



Python + FastAPI

Backend framework and API development



MongoDB

Database for storing user data and learning content



Google Gemini AI

Generates summaries, flashcards, quizzes and study plans



JWT

Secure user authentication using tokens



CORS

Enables frontend-backend communication



Environment Variables

API keys, database URI and sensitive configs



Testing Tools



Postman

- Test API endpoints
- Check request/response data
- Validate status codes



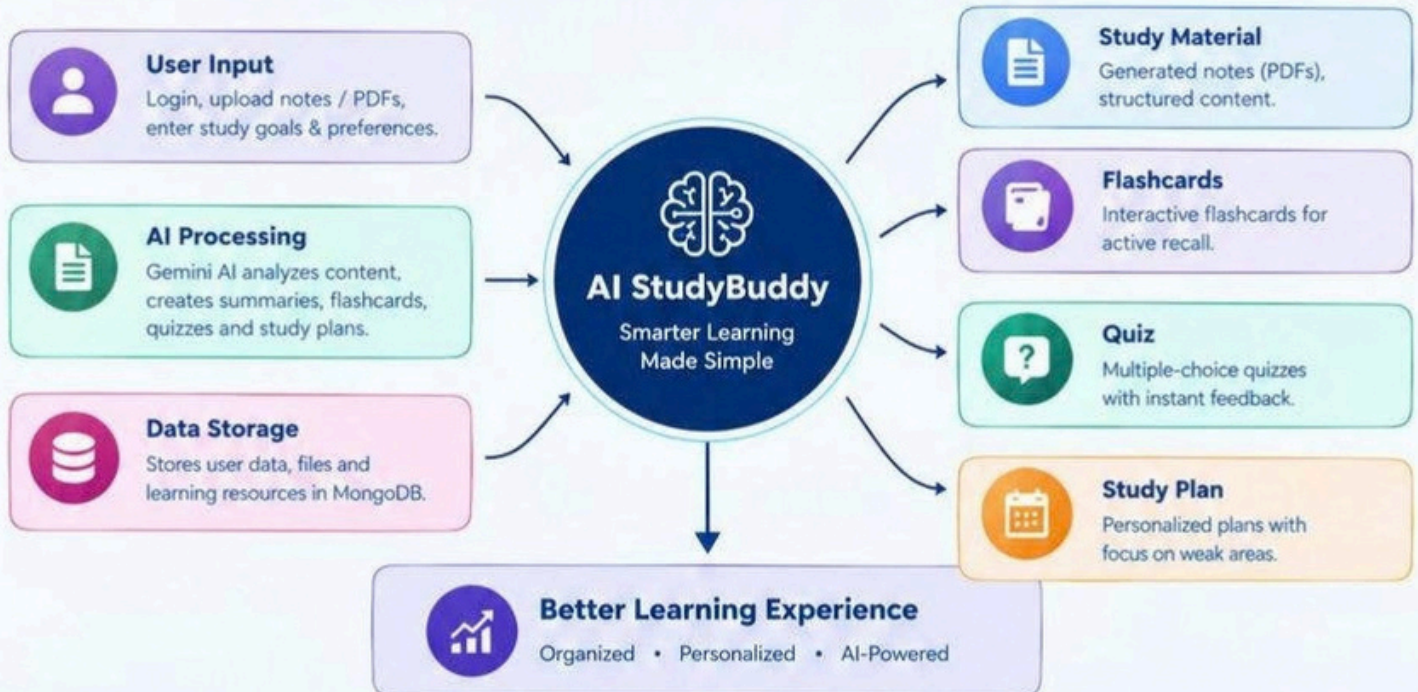
Result

All endpoints working as expected with correct data flow and error handling.



Project Outcome

AI StudyBuddy successfully helps students manage their learning, generate study content and stay consistent with their goals using AI and modern web technologies.



Key Benefits



Personalized Learning

Adapts to user goals and study patterns for better results.



Saves Time

Generates summaries, flashcards and quizzes instantly.



Improves Retention

Interactive and structured content helps in long-term memory.



Tracks Progress

Monitors performance and highlights weak areas.



Easy to Use

Simple and clean interface with secure access.



Accessible Anytime

Learn from anywhere, anytime with an internet connection.



Future Enhancements



AI Digital Twin Tutor

Personalized virtual tutor with real-time feedback and guidance.



AR / MR Learning

3D & interactive learning experiences for better understanding.



Emotion & Focus Detection

Track mood and attention using smart sensors (e.g., webcam/wearables).



Predictive Analytics

Predict performance & suggest study actions using data trends.



BCI Integration

Brain-computer interface for next-level learning (future research).



More Content & Features

Add more subjects, gamification, community and multi-language support.



DEMO LINK :

https://drive.google.com/file/d/1erJ66xGWI4_QSaZy1MLcwNBuR9DbQ8X8/view?usp=drivesdk

